PIZZA SALES ANALYSIS USING SQL

PROJECT OVERVIEW :

This project focuses on the analysis of pizza sales data to gain insights into customer preferences, revenue generation, and order patterns.

The dataset used in this analysis contains information about pizza sales during the year 2015. This project analyzes a pizza sales dataset using SQL queries.

By examining the data, my aim is to uncover valuable insights that can help inform business decisions and optimize the pizza sales operations.

Tools & Technologies Used :

- MySQL
- SQL Server Management Studio (SSMS)

OBJECTIVES :

- Analyze total revenue generated from pizza sales
- Identify the most popular pizzas
- Understand sales distribution by pizza category and size
- Analyze monthly order trends
- Identify top revenue-generating pizzas

BUSINESS POTENTIAL :

Understanding customer preferences and market trends is crucial for businesses to thrive in a competitive environment. Analyzing sales data allows us to gain a deeper understanding of customer behavior, identify popular pizza choices, and recognize patterns in customer orders.

DATASET DESCRIPTION:

Column Name	Description
pizza_id	Unique identifier for each pizza record
order_id	Unique identifier for each order
pizza_name_id	Identifier for pizza type
quantity	Number of pizzas ordered
order_date	Date when the order was placed
order_time	Time when the order was placed
unit_price	Price of one pizza
total_price	Total price for the order
pizza_size	Size of pizza (S, M, L, XL, XXL)
pizza_category	Category of pizza (Classic, Chicken, Veggie, Supreme)
pizza_ingredients	Ingredients used
pizza_name	Name of the pizza

BASIC QUERIES:

1. **What is the total revenue generated from all pizza sales?**

```
SELECT ROUND(SUM(total_price), 2) AS total_revenue
FROM pizza_sales;
```

	total_revenue
1	817860.05

2. How many total distinct orders were placed?

```
SELECT COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales;
```

	total_orders
1	21350

3. What is the total number of pizzas sold?

```sql
SELECT SUM(quantity) AS total_pizzas_sold
FROM pizza_sales;
```

	total_pizzas_sold
1	49574

4. What is the average order value of all pizzas?

```sql
SELECT ROUND(SUM(total_price) / COUNT(DISTINCT order_id), 2)
AS avg_order_value
FROM pizza_sales;
```

	avg_order_value
1	38.31

5. What is the max and min order value?

```
SELECT MAX(total_price) AS max_price, MIN (total_price) AS min_price
FROM pizza_sales;
```

	max_price	min_price
1	83	9.75

INTERMEDIATE QUERIES:

6. What is the total revenue generated by each pizza category?

```
SELECT pizza_category, SUM(total_price) AS total_revenue
FROM pizza_sales
GROUP BY pizza_category;
```

	pizza_category	total_revenue
1	Classic	220053.1
2	Chicken	195919.5
3	Veggie	193690.45
4	Supreme	208197

7. What is the total quantity sold for each pizza size ordered descending?

```
SELECT pizza_size, SUM(quantity) AS total_quantity
FROM pizza_sales
GROUP BY pizza_size
ORDER BY total_quantity DESC;
```

	pizza_size	total_quantity
1	L	18956
2	M	15635
3	S	14403
4	XL	552
5	XXL	28

8. Which pizza generated the highest revenue?

```sql
SELECT TOP 1 pizza_name, SUM(total_price) AS revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY revenue DESC;
```

	pizza_name	revenue
1	The Thai Chicken Pizza	43434.25

9. What is the total numbers of orders placed per month?

```sql
SELECT DATENAME(MONTH, order_date) AS order_month,
       COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY DATENAME(MONTH, order_date), MONTH(order_date)
ORDER BY MONTH(order_date);
```

	order_month	total_orders
1	January	1845
2	February	1685
3	March	1840
4	April	1799
5	May	1853
6	June	1773
7	July	1935
8	August	1841
9	September	1661
10	October	1646
11	November	1792
12	December	1680

10. What are the top 5 best-selling pizzas based on quantity sold?

```sql
SELECT TOP 5 pizza_name, SUM(quantity) AS total_quantity
FROM pizza_sales
GROUP BY pizza_name
ORDER BY total_quantity DESC;
```

	pizza_name	total_quantity
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Pizza	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

ADVANCED QUERIES:

11. Find the second highest revenue-generating pizza using a subquery or window function

```
SELECT pizza_name, revenue FROM
(
  SELECT pizza_name,
             SUM(total_price) AS revenue,
             DENSE_RANK() OVER (ORDER BY SUM(total_price) DESC) AS rank_num
      FROM pizza_sales
      GROUP BY pizza_name
  ) t
WHERE rank_num = 2;
```

	pizza_name	revenue
1	The Barbecue Chicken Pizza	42768

12. Which top 5 pizzas generate revenue higher than the average pizza revenue?

```sql
SELECT TOP 5 pizza_name, SUM(total_price) AS revenue
FROM pizza_sales
GROUP BY pizza_name
HAVING SUM(total_price) > (SELECT AVG(revenue) FROM
(
   SELECT SUM(total_price) AS revenue
     FROM pizza_sales
    GROUP BY pizza_name
))
ORDER BY revenue DESC;
```

	pizza_name	revenue
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Pizza	42768
3	The California Chicken Pizza	41409.5
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

INSIGHTS:

- The business generated **$817,860.05 in total revenue** from 21,350 orders, indicating strong overall sales performance.
- A total of **49,574 pizzas were sold**, showing high customer demand for pizza products.
- **Classic pizzas generated the highest revenue**, making them the most profitable category.
- **Large pizzas are the most popular size**, with the highest quantity sold among all sizes.
- **The Thai Chicken Pizza and Barbecue Chicken Pizza** are among the top revenue-generating pizzas.

RECOMMENDATIONS:

- Promote top-performing pizzas such as Thai Chicken and Barbecue Chicken pizzas to increase sales.
- Introduce combo offers on large pizzas, as they are the most preferred size.
- Run seasonal promotions during low-demand months to maintain consistent sales.
- Expand the Classic pizza category with new flavors since it generates the highest revenue.
- Use sales insights to create data-driven marketing strategies and improve menu planning.

CONCLUSION:

The SQL analysis of the pizza sales dataset provided valuable insights into revenue performance, customer preferences, and product demand. The business generated strong revenue with a high number of orders and pizzas sold throughout the year.

The analysis highlighted that Classic pizzas and large pizza sizes contribute significantly to sales, while certain pizzas such as Thai Chicken and Barbecue Chicken play a major role in driving revenue.

These insights can help the business make data-driven decisions, improve marketing strategies, and optimize menu offerings to enhance overall sales performance and customer satisfaction.

THANK YOU